

Quantum Information Theory

Lesson 5: Quantum Channels

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Remember: Classical Information

The state of a d -level classical system is an $x \in \{0, 1, \dots, d-1\}$. If the system state is a random variable $X : \Omega \rightarrow \{0, 1, \dots, d-1\}$, we can model the probabilities of each state as a vector $\mathbf{p} \in [0, 1]^d$,

$$p_i = \Pr[X = i].$$

$\mathbf{p} \in \mathbb{R}^d$ is probabilistic iff its entries are non-negative and sum to 1, and we model uncertainty of X using $H(X) = -\sum_{i=0}^{d-1} p_i \log p_i$.

If we transform X into Y using a Markovian (memory-less) process,

$$q_j = \Pr[Y = j] = \sum_{i=0}^{d-1} \Pr[Y = j \mid X = i] \Pr[X = i] = (\mathbf{D}\mathbf{p})_j,$$

where the matrix $\mathbf{D}$ has entries $d_{j,i} = \Pr[Y = j \mid X = i]$.

$\mathbf{D}$ map probability vectors to probability vectors if and only if

$$\frac{1}{d} [1 \quad 1 \quad \dots \quad 1] \mathbf{D} |i\rangle = \frac{1}{d} \quad \forall i$$

so we say matrix $\mathbf{D}$ is stochastic iff all its columns sum to 1.

Remember: Classical Information

After applying an arbitrary $\mathbf{D}$, entropy may increase or decrease.

For $\mathbf{D}$ to never increase entropy, it must map all $|i\rangle$ into some $|j\rangle$ so its entries are all 0 or 1, so $Y = f(X)$ for some deterministic f . But we always have $H(f(X)) \leq H(X)$ so we find the equivalence:

Classical channel $\mathbf{D}$ can't increase entropy iff all $d_{j,i} \in \{0, 1\}$.

For $\mathbf{D}$ to never decrease entropy, it must map $\frac{1}{d} [1 \ 1 \ \dots \ 1]^T$ to itself hence $\langle i | \mathbf{D} \frac{1}{d} [1 \ 1 \ \dots \ 1]^T = \frac{1}{d}$ i.e. $\mathbf{D}^\dagger$ must also be stochastic, which says that $\mathbf{D}$'s rows should also sum to 1, making $\mathbf{D}$ doubly stochastic (also known as bistochastic).

But if $\mathbf{D}$ is doubly stochastic then $\mathbf{p} \succeq \mathbf{q}$ so $H(\mathbf{q}) \geq H(\mathbf{p})$, thus

$\mathbf{D}$ can't decrease entropy iff $\mathbf{D} \frac{1}{d} [1 \ \dots \ 1]^T = \frac{1}{d} [1 \ \dots \ 1]^T$.

But, even when $\mathbf{D}$ can do both, it always shrinks KL -divergence:

$$D_{KL}(\mathbf{p}_1 \| \mathbf{p}_2) \geq D_{KL}(\mathbf{D}\mathbf{p}_1 \| \mathbf{D}\mathbf{p}_2),$$

something we call the Data Processing Inequality.

Remember: Quantum Information

The state of a d -level quantum system is a $|\psi\rangle \in \mathbb{C}^d$.

If the system state is a random variable $|\mathbf{v}\rangle : \Omega \rightarrow \mathbb{C}^d$, with

$$|\mathbf{v}\rangle \sim \begin{pmatrix} |\psi_1\rangle & |\psi_2\rangle & \dots & |\psi_n\rangle \\ p_1 & p_2 & \dots & p_n \end{pmatrix},$$

we model it as a density matrix $\rho = \sum_{i=1}^n p_i |\psi_i\rangle \langle \psi_i| \in \mathcal{B}(\mathbb{C}^d)$.

$\rho \in \mathcal{B}(\mathbb{C}^d)$ is a density matrix iff $\rho = \rho^\dagger$, $\rho \geq 0$ and $\text{Tr } \rho = 1$,
and we model uncertainty of ρ using $H(\rho) = -\text{Tr } \rho \log \rho$.

Many quantum random variables have the same density matrix.
In particular, the spectral decomposition $\rho = \sum_{i=1}^d \lambda_i |\phi_i\rangle \langle \phi_i|$
indicates that ρ is associated to the quantum random variable

$$|\mathbf{v}\rangle \sim \begin{pmatrix} |\phi_1\rangle & |\phi_2\rangle & \dots & |\phi_d\rangle \\ \lambda_1 & \lambda_2 & \dots & \lambda_d \end{pmatrix},$$

and we can calculate $H(\rho) = -\text{Tr } \rho \log \rho = -\sum_{i=1}^d \lambda_i \log \lambda_i$.

Quantum Channels

If we transform $|x\rangle$ to $|y\rangle$ using a Markovian (memory-less) process, let's say a function $\mathcal{E} : \mathcal{S}(\mathbb{C}^d) \rightarrow \mathcal{S}(\mathbb{C}^{d'})$ we find

$$\mathbb{E}[|y\rangle\langle y|] = \int \mathbb{E}[|y\rangle\langle y| \mid |x\rangle = |\psi\rangle] \Pr[|x\rangle = |\psi\rangle] d|\psi\rangle,$$

so in particular $\mathcal{E}(p\rho + (1-p)\rho') = p\mathcal{E}(\rho) + (1-p)\mathcal{E}(\rho')$,
for any density matrices ρ, ρ' and any $0 \leq p \leq 1$,
so $\mathcal{E}$'s restriction to density matrices is linear.

Looking at the basis of $\mathcal{B}(\mathbb{C}^d)$ comprised of the density matrices,

$$\mathbf{Z}_a = |a\rangle\langle a|, \quad \mathbf{X}_{b,c} = \frac{|b\rangle + |c\rangle}{\sqrt{2}} \frac{\langle b| + \langle c|}{\sqrt{2}}, \quad \mathbf{Y}_{d,e} = \frac{|d\rangle + i|e\rangle}{\sqrt{2}} \frac{\langle d| - i\langle e|}{\sqrt{2}},$$

we find that any matrix $\mathbf{A} = \sum_{i,j=0}^{d-1} a_{i,j} |i\rangle\langle j|$ can be expressed as

$$\mathbf{A} = \sum_{i=0}^{d-1} a_{i,i} \mathbf{Z}_i + \sum_{0 \leq i < j < d} (a_{i,j} + a_{j,i}) \left(\mathbf{X}_{i,j} - \frac{\mathbf{Z}_i + \mathbf{Z}_j}{2} \right) + i \sum_{0 \leq i < j < d} (a_{i,j} - a_{j,i}) \left(\mathbf{Y}_{i,j} - \frac{\mathbf{Z}_i + \mathbf{Z}_j}{2} \right)$$

so there is a unique linear extension of $\mathcal{E}$ to the entire $\mathcal{B}(\mathbb{C}^d)$.

Quantum Channels

A quantum channel is a linear map between the mixed state spaces of two systems $\mathcal{E} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H}')$ s.t. its action on (part of) a system maps density matrices to density matrices.

We now want a more formal definition of quantum channels, like we had for classical channels, which were stochastic matrices.

A linear map $\mathcal{E} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H}')$ is said to be

- ▶ **positive**: if $\mathcal{E}(\mathbf{A}) \geq 0 \ \forall \mathbf{A} \geq 0$;
- ▶ **n -positive**: if $\mathbf{I}_n \otimes \mathcal{E}(\mathbf{A}) \geq 0 \ \forall \mathbf{A} \geq 0$, where $n \in \{1, 2, 3, \dots\}$;
- ▶ **completely positive**: if $\mathbf{I}_n \otimes \mathcal{E}(\mathbf{A}) \geq 0 \ \forall \mathbf{A} \geq 0 \ \forall n \in \mathbb{N}^*$;
- ▶ **trace preserving**: if $\text{Tr}(\mathcal{E}(\mathbf{A})) = \text{Tr} \mathbf{A} \ \forall \mathbf{A}$;
- ▶ **unital**: if $\mathcal{E}(\mathbf{I}_{\mathcal{H}}) = \mathbf{I}_{\mathcal{H}'}$;
- ▶ **unitary**: if $\mathcal{E}(\mathbf{A}) = \mathbf{U} \mathbf{A} \mathbf{U}^\dagger \ \forall \mathbf{A}$ for some unitary matrix $\mathbf{U}$;
- ▶ **self-adjoint**: if $\mathcal{E} = \mathcal{E}^\dagger$ where the dual channel $\mathcal{E}^\dagger$ is given by $\text{Tr}((\mathcal{E}^\dagger(\mathbf{A}))^\dagger \mathbf{B}) = \text{Tr}(\mathbf{A}^\dagger \mathcal{E}(\mathbf{B})) \ \forall \mathbf{A} \in \mathcal{B}(\mathcal{H}'), \mathbf{B} \in \mathcal{B}(\mathcal{H})$.

Quantum Channels

Note that for any matrix $\mathbf{A} \in \mathcal{B}(\mathcal{H})$ we have $\mathbf{A} = \frac{\mathbf{A} + \mathbf{A}^\dagger}{2} + \frac{\mathbf{A} - \mathbf{A}^\dagger}{2}$, where the first term is Hermitian and the second is anti-Hermitian. Yet, any Hermitian matrix is the sum of a positive semi-definite matrix and a negative semi-definite matrix. So, we can write

$$\mathbf{A} = \alpha_1 \rho_1 - \alpha_2 \rho_2 + i\alpha_3 \rho_3 - i\alpha_4 \rho_4.$$

First, we find that quantum channels must be trace preserving:

$$\begin{aligned}\text{Tr}(\mathcal{E}(\mathbf{A})) &= \alpha_1 \text{Tr}(\mathcal{E}(\rho_1)) - \alpha_2 \text{Tr}(\mathcal{E}(\rho_2)) + i\alpha_3 \text{Tr}(\mathcal{E}(\rho_3)) - i\alpha_4 \text{Tr}(\mathcal{E}(\rho_4)) \\ &= \alpha_1 - \alpha_2 + i\alpha_3 - i\alpha_4 = \text{Tr} \mathbf{A}.\end{aligned}$$

Next, it follows that they must be completely positive:

$$\mathbf{I}_n \otimes \mathcal{E}(\alpha_1 \rho_1) = \alpha_1 (\mathbf{I}_n \otimes \mathcal{E}(\rho_1)) \geq 0.$$

So, the more formal (but equivalent) definition is that:

Quantum channels are completely positive trace preserving maps.
(CPTP maps)

Quantum Channels

The first thing we would like to be a channel is quantum gates,

$$\mathcal{E}(\rho) = \mathcal{E}\left(\sum_i p_i |\psi_i\rangle \langle \psi_i|\right) = \sum_i p_i \mathcal{E}(|\psi_i\rangle \langle \psi_i|) = \sum_i p_i \mathbf{U} |\psi_i\rangle \langle \psi_i| \mathbf{U}^\dagger = \mathbf{U} \rho \mathbf{U}^\dagger.$$

The second thing is random quantum gates:

$$\mathcal{E}(\rho) = \sum_i p_i \mathbf{U}_i \rho \mathbf{U}_i^\dagger = \sum_i (\sqrt{p_i} \mathbf{U}_i) \rho (\sqrt{p_i} \mathbf{U}_i)^\dagger,$$

so we look at linear maps that can be written to look like

$$\mathcal{E}(\rho) = \sum_{k=0}^{m-1} \mathbf{K}_k \rho \mathbf{K}_k^\dagger.$$

Maps $\mathcal{E}$ as above are said to admit a Kraus representation, and the matrices $\mathbf{K}_k \in \text{Mat}_{d',d}(\mathbb{C})$ are called Kraus operators. A Kraus representation is called complete if $\sum_{k=0}^{m-1} \mathbf{K}_k^\dagger \mathbf{K}_k = \mathbf{I}_d$.

Kraus Representation $\Rightarrow$ Completely Positive

$$\begin{aligned}
 \mathbf{I}_n \otimes \mathcal{E} \left(\begin{bmatrix} \mathbf{A}_{0,0} & \cdots & \mathbf{A}_{0,n-1} \\ \vdots & \ddots & \vdots \\ \mathbf{A}_{n-1,0} & \cdots & \mathbf{A}_{n-1,n-1} \end{bmatrix} \right) &= \begin{bmatrix} \mathcal{E}(\mathbf{A}_{0,0}) & \cdots & \mathcal{E}(\mathbf{A}_{0,n-1}) \\ \vdots & \ddots & \vdots \\ \mathcal{E}(\mathbf{A}_{n-1,0}) & \cdots & \mathcal{E}(\mathbf{A}_{n-1,n-1}) \end{bmatrix} \\
 &= \begin{bmatrix} \sum_{k=0}^{m-1} \mathbf{K}_k \mathbf{A}_{0,0} \mathbf{K}_k^\dagger & \cdots & \sum_{k=0}^{m-1} \mathbf{K}_k \mathbf{A}_{0,n-1} \mathbf{K}_k^\dagger \\ \vdots & \ddots & \vdots \\ \sum_{k=0}^{m-1} \mathbf{K}_k \mathbf{A}_{n-1,0} \mathbf{K}_k^\dagger & \cdots & \sum_{k=0}^{m-1} \mathbf{K}_k \mathbf{A}_{n-1,n-1} \mathbf{K}_k^\dagger \end{bmatrix} \\
 &= \sum_{k=0}^{m-1} \begin{bmatrix} \mathbf{K}_k \mathbf{A}_{0,0} \mathbf{K}_k^\dagger & \cdots & \mathbf{K}_k \mathbf{A}_{0,n-1} \mathbf{K}_k^\dagger \\ \vdots & \ddots & \vdots \\ \mathbf{K}_k \mathbf{A}_{n-1,0} \mathbf{K}_k^\dagger & \cdots & \mathbf{K}_k \mathbf{A}_{n-1,n-1} \mathbf{K}_k^\dagger \end{bmatrix} \\
 &= \sum_{k=0}^{m-1} \begin{bmatrix} \mathbf{K}_k & & \\ & \ddots & \\ & & \mathbf{K}_k \end{bmatrix} \begin{bmatrix} \mathbf{A}_{0,0} & \cdots & \mathbf{A}_{0,n-1} \\ \vdots & \ddots & \vdots \\ \mathbf{A}_{n-1,0} & \cdots & \mathbf{A}_{n-1,n-1} \end{bmatrix} \begin{bmatrix} \mathbf{K}_k^\dagger & & \\ & \ddots & \\ & & \mathbf{K}_k^\dagger \end{bmatrix} \\
 &= \sum_{k=0}^{m-1} (\mathbf{I} \otimes \mathbf{K}_k) \mathbf{A} (\mathbf{I} \otimes \mathbf{K}_k^\dagger) = \sum_{k=0}^{m-1} \underbrace{(\mathbf{I} \otimes \mathbf{K}_k) \sqrt{\mathbf{A}}}_{\mathbf{B}_k} \underbrace{\sqrt{\mathbf{A}}^\dagger (\mathbf{I} \otimes \mathbf{K}_k^\dagger)}_{\mathbf{B}_k^\dagger} \geq 0
 \end{aligned}$$

d -positive $\Rightarrow$ Choi Matrix ≥ 0

Let's now take a generic $\mathcal{E} : \mathcal{B}(\mathbb{C}^d) \rightarrow \mathcal{B}(\mathbb{C}^{d'})$ and see what it does to a Bell state $|\beta_{00}\rangle = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |i\rangle \otimes |i\rangle$:

$$\begin{aligned} \mathbf{I}_d \otimes \mathcal{E} (|\beta_{00}\rangle \langle \beta_{00}|) &= \mathbf{I}_d \otimes \mathcal{E} \left(\frac{1}{d} \sum_{i,j=0}^{d-1} |i\rangle \langle j| \otimes |i\rangle \langle j| \right) \\ &= \frac{1}{d} \sum_{i,j=0}^{d-1} |i\rangle \langle j| \otimes \mathcal{E}(|i\rangle \langle j|) \\ &= \frac{1}{d} \begin{bmatrix} \mathcal{E}(|0\rangle \langle 0|) & \dots & \mathcal{E}(|0\rangle \langle d-1|) \\ \vdots & \ddots & \vdots \\ \mathcal{E}(|d-1\rangle \langle 0|) & \dots & \mathcal{E}(|d-1\rangle \langle d-1|) \end{bmatrix} = \mathbf{J}_{\mathcal{E}}, \end{aligned}$$

a positive semi-definite matrix known as $\mathcal{E}$'s **Choi matrix**.

Note: $\mathcal{E} \leftrightarrow \mathbf{J}_{\mathcal{E}} = \mathbf{I} \otimes \mathcal{E}(|\beta_{00}\rangle \langle \beta_{00}|)$ is called the **Jamiołkowski isomorphism** (between channels and states).

Choi Matrix $\geq 0 \Rightarrow$ Kraus Representation

Any $dd' \times dd'$ matrix can be written as a block matrix.

From it we extract the $d' \times d'$ blocks that would be $\mathcal{E}(|i\rangle \langle j|)$.

And so, from any $dd' \times dd'$ matrix we can make a linear $\mathcal{E}$.

Now, let's take a density matrix $\mathbf{M} \in \mathcal{B}(\mathbb{C}^{dd'})$ with spectral decomposition $\mathbf{M} = \sum_{k=0}^{r-1} \lambda_k |\psi_k\rangle \langle \psi_k|$ and let $|\psi'_k\rangle = \sqrt{\lambda_k} |\psi_k\rangle$.

$$\begin{aligned} \mathcal{E}(|i\rangle \langle j|) &= d(\langle i| \otimes \mathbf{I}_{d'}) \mathbf{M} (|j\rangle \otimes \mathbf{I}_{d'}) \\ &= \sum_{k=0}^{r-1} \left(\sqrt{d}(\langle i| \otimes \mathbf{I}_{d'}) |\psi'_k\rangle \right) \left(\sqrt{d}(\langle j| \otimes \mathbf{I}_{d'}) |\psi'_k\rangle \right)^\dagger, \end{aligned}$$

and notice that taking $\boxed{\mathbf{K}_k = \sqrt{d} \sum_{a=0}^{d-1} (\langle a| \otimes \mathbf{I}_{d'}) |\psi'_k\rangle \langle a|}$ we find

$$\mathcal{E}(|i\rangle \langle j|) = \sum_{k=0}^{r-1} \mathbf{K}_k |i\rangle \langle j| \mathbf{K}_k^\dagger.$$

From the $\mathcal{E}(|i\rangle \langle j|)$ we get

$$\mathcal{E}(\rho) = \sum_{i,j=0}^{d-1} \langle i| \rho |j\rangle \mathcal{E}(|i\rangle \langle j|) = \sum_{i,j=0}^{d-1} \langle i| \rho |j\rangle \sum_{k=0}^{r-1} \mathbf{K}_k |i\rangle \langle j| \mathbf{K}_k^\dagger = \sum_{k=0}^{r-1} \mathbf{K}_k \rho \mathbf{K}_k^\dagger.$$

and, as we have seen, any map of this form is completely positive. 12 / 30

Choi's Theorem on completely positive maps

Let $\mathcal{E} : \mathcal{B}(\mathbb{C}^d) \rightarrow \mathcal{B}(\mathbb{C}^{d'})$ be a linear map.

The following statements are equivalent:

1. $\mathcal{E}$ has a representation $\mathcal{E}(\rho) = \sum_{k=0}^{m-1} \mathbf{K}_k \rho \mathbf{K}_k^\dagger$ for some m ;
2. $\mathcal{E}$ is completely positive;
3. $\mathcal{E}$ is d -positive;
4. $\mathcal{E}$'s Choi matrix $\mathbf{J}_\mathcal{E} = \mathbf{I}_d \otimes \mathcal{E}(|\beta_{00}\rangle \langle \beta_{00}|) \geq 0$;
5. $\mathcal{E}$ has a Kraus representation $\mathcal{E}(\rho) = \sum_{k=0}^{\text{rank } \mathbf{J}_\mathcal{E}-1} \mathbf{K}_k \rho \mathbf{K}_k^\dagger$.

Note: For any Kraus representation $\mathcal{E} = \sum_{k=0}^{m-1} \mathbf{K}_k \rho \mathbf{K}_k^\dagger$ we have $m \geq \text{rank } \sum_{k=0}^{m-1} (\mathbf{I}_d \otimes \mathbf{K}_k) |\beta_{00}\rangle \langle \beta_{00}| ((\mathbf{I}_d \otimes \mathbf{K}_k^\dagger)) = \text{rank } \mathbf{J}_\mathcal{E}$.

Corollary: $\mathcal{E} : \mathcal{B}(\mathbb{C}^d) \rightarrow \mathcal{B}(\mathbb{C}^d)$ is unitary iff $\text{rank } \mathbf{J}_\mathcal{E} = 1$.

$$\Rightarrow: (\mathbf{I}_d \otimes \mathcal{E})(|\beta_{00}\rangle \langle \beta_{00}|) = \underbrace{(\mathbf{I}_d \otimes \mathbf{U})}_{|\psi\rangle} |\beta_{00}\rangle \underbrace{\langle \beta_{00}| (\mathbf{I}_d \otimes \mathbf{U}^\dagger)}_{\langle \psi|}$$

$\Leftarrow$: If $\mathbf{J}_\mathcal{E}$ has rank 1 then $\mathcal{E}(\rho) = \mathbf{K} \rho \mathbf{K}^\dagger$ and

$$\delta_{i,j} = \text{Tr } |i\rangle \langle j| = \text{Tr}(\mathcal{E}(|i\rangle \langle j|)) = \text{Tr}(\mathbf{K} |i\rangle \langle j| \mathbf{K}^\dagger) = \langle j| \mathbf{K}^\dagger \mathbf{K} |i\rangle,$$

i.e. $\mathbf{K}^\dagger \mathbf{K} = \mathbf{I}$ and all square isometries are unitary.

Example Channels

For the completely depolarizing map $\Delta_1(\rho) = \frac{\mathbf{I}_d}{d} \text{Tr } \rho$ we find

$$\mathbf{J}_{\Delta_1} = \mathbf{I}_d \otimes \Delta_1(|\beta_{00}\rangle \langle \beta_{00}|) = \frac{1}{d} \begin{bmatrix} \frac{\mathbf{I}_d}{d} & & \\ & \ddots & \\ & & \frac{\mathbf{I}_d}{d} \end{bmatrix} = \frac{\mathbf{I}_{d^2}}{d^2} \geq 0$$

and $\text{rank } \mathbf{J}_{\Delta_1} = d^2$ so a minimal Kraus representation has d^2 operators. We find the orthogonal eigenvectors $|k_1\rangle \otimes |k_2\rangle$ for $k_1, k_2 \in \{0, \dots, d-1\}$ corresponding to eigenvalue $\lambda = 1/d^2$, so

$$\mathbf{K}_{k_1, k_2} = \sqrt{d} \sum_{a=0}^{d-1} (\langle a| \otimes \mathbf{I}_d) \left(\sqrt{\frac{1}{d^2}} |k_1\rangle \otimes |k_2\rangle \right) \langle a| = \frac{1}{\sqrt{d}} |k_2\rangle \langle k_1|,$$

is a minimal Kraus representation. Since $\mathbf{K}_{i_1, i_2}^\dagger \mathbf{K}_{j_1, j_2} = \frac{\delta_{i_2, j_2}}{d} |i_1\rangle \langle j_1|$ the Kraus operators are orthogonal and $\sum_{k_1, k_2=0}^{d-1} \mathbf{K}_{k_1, k_2}^\dagger \mathbf{K}_{k_1, k_2} = \mathbf{I}_d$.

Example Channels

Note that this decomposition is not unique. Computing

$$\begin{aligned}
 & \frac{1}{d^2} \sum_{k_1, k_2} \mathbf{z}_d^{k_2} \mathbf{x}_d^{k_1} \rho \mathbf{x}_d^{k_1 \dagger} \mathbf{z}_d^{k_2 \dagger} = \frac{1}{d^2} \sum_{k_1, k_2, i, j} \mathbf{z}_d^{k_2} \mathbf{x}_d^{k_1} |i\rangle \langle i| \rho |j\rangle \langle j| \mathbf{x}_d^{k_1 \dagger} \mathbf{z}_d^{k_2 \dagger} \\
 &= \frac{1}{d^2} \sum_{k_1, k_2, i, j} \mathbf{z}_d^{k_2} |i \oplus k_1\rangle \langle i| \rho |j\rangle \langle j \oplus k_1| \mathbf{z}_d^{k_2 \dagger} \\
 &= \frac{1}{d^2} \sum_{k_1, k_2, i, j} e^{\frac{2\pi i k_2(i+k_1)}{d}} |i \oplus k_1\rangle \langle i| \rho |j\rangle \langle j \oplus k_1| e^{-\frac{2\pi i k_2(j+k_1)}{d}} \\
 &= \frac{1}{d^2} \sum_{k_1, i, j} \underbrace{\left(\sum_{k_2} e^{\frac{2\pi i k_2(i-j)}{d}} \right)}_{d\delta_{i,j}} |i \oplus k_1\rangle \langle i| \rho |j\rangle \langle j \oplus k_1| \\
 &= \frac{1}{d} \sum_{k_1, i} |i \oplus k_1\rangle \langle i| \rho |i\rangle \langle i \oplus k_1| = \frac{1}{d} \sum_{k_1, i} |i\rangle \langle i \ominus k_1| \rho |i \ominus k_1\rangle \langle i| = \frac{1}{d} \text{Tr } \rho
 \end{aligned}$$

we find this is another minimal Kraus representation of Δ_1 .

Example Channels

For the 2D depolarizing map $\Delta_\epsilon(\rho) = (1 - \epsilon)\rho + \epsilon \frac{\mathbf{I}_2}{2} \text{Tr } \rho$ we find

$$\mathbf{J}_{\Delta_\epsilon} = \mathbf{I}_2 \otimes \Delta_1(|\beta_{00}\rangle \langle \beta_{00}|) = \frac{1}{2} \begin{bmatrix} 1 - \epsilon/2 & & & 1 - \epsilon \\ & \epsilon/2 & & \\ & & \epsilon/2 & \\ 1 - \epsilon & & & 1 - \epsilon/2 \end{bmatrix} \geq 0$$

and $\text{rank } \mathbf{J}_{\Delta_\epsilon} = 4$ so a minimal Kraus representation has 4 operators. From the orthogonal eigenvectors we find

$$\lambda_0 = \frac{4 - 3\epsilon}{4}; \quad |\psi_0\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle; \quad \mathbf{K}_0 = \frac{\sqrt{4 - 3\epsilon}}{2} \mathbf{I}_2;$$

$$\lambda_1 = \frac{\epsilon}{4}; \quad |\psi_1\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle; \quad \mathbf{K}_1 = \frac{\sqrt{\epsilon}}{2} \mathbf{Z};$$

$$\lambda_2 = \frac{\epsilon}{4}; \quad |\psi_2\rangle = |01\rangle; \quad \mathbf{K}_2 = \sqrt{\frac{\epsilon}{2}} |1\rangle \langle 0|;$$

$$\lambda_3 = \frac{\epsilon}{4}; \quad |\psi_3\rangle = |10\rangle; \quad \mathbf{K}_3 = \sqrt{\frac{\epsilon}{2}} |0\rangle \langle 1|;$$

$$\sum_{k=0}^3 \mathbf{K}_k \rho \mathbf{K}_k^\dagger = \frac{4 - 3\epsilon}{4} \rho + \frac{\epsilon}{4} \begin{bmatrix} \rho_{00} & -\rho_{01} \\ -\rho_{10} & \rho_{11} \end{bmatrix} + \frac{\epsilon}{4} \begin{bmatrix} \rho_{11} & 0 \\ 0 & 0 \end{bmatrix} + \frac{\epsilon}{4} \begin{bmatrix} 0 & 0 \\ 0 & \rho_{00} \end{bmatrix} = (1 - \epsilon)\rho + \frac{\epsilon \mathbf{I}_2 \text{Tr } \rho}{2}.$$

Example Channels

- ▶ A gate $\mathbf{U}$ is a Kraus channel with operators $\{\mathbf{U}\}$

$$\mathcal{E}(\rho) = \mathbf{U}\rho\mathbf{U}^\dagger;$$

- ▶ Adding an ancilla is a Kraus channel with operators $\{\mathbf{I} \otimes |0\rangle\}$

$$\mathcal{E}(\rho) = \rho \otimes |0\rangle\langle 0| = (\mathbf{I} \otimes |0\rangle\langle 0|)\rho(\mathbf{I} \otimes \langle 0|);$$

- ▶ Partial trace is a Kraus channel with operators $\{\mathbf{I} \otimes \langle i|\}$:

$$\mathrm{Tr}_B(\rho_{AB}) = \sum_{i=0}^{\dim \mathcal{H}_B - 1} \mathbf{I} \otimes \langle i| \rho_{AB} \mathbf{I} \otimes |i\rangle$$

- ▶ Measuring $\mathbf{O} = \sum_\lambda \lambda E_\lambda(\mathbf{O})$ and recording the result is a Kraus channel with operators $\{E_\lambda(\mathbf{O}) \otimes |\lambda\rangle\}$

$$\mathcal{E}(\rho) = \sum_\lambda \mathrm{Tr}(E_\lambda(\mathbf{O})\rho E_\lambda(\mathbf{O})) \frac{E_\lambda(\mathbf{O})\rho E_\lambda(\mathbf{O}) \otimes |\lambda\rangle\langle \lambda|}{\mathrm{Tr}(E_\lambda(\mathbf{O})\rho E_\lambda(\mathbf{O}))}$$

Partial Trace

Let's take the block matrix $\mathbf{A} \in \mathcal{B}(\mathbb{C}^d \otimes \mathbb{C}^{d'})$,

$$\begin{aligned}\mathbf{A} &= \begin{bmatrix} \mathbf{A}_{0,0} & \dots & \mathbf{A}_{0,d-1} \\ \vdots & \ddots & \vdots \\ \mathbf{A}_{d-1,0} & \dots & \mathbf{A}_{d-1,d-1} \end{bmatrix} \\ &= \sum_{i,j} |i\rangle \langle j| \otimes \mathbf{A}_{i,j}\end{aligned}$$

and notice that tracing out the halves does different things,

$$\begin{aligned}\mathrm{Tr}_{\mathbb{C}^d}(\mathbf{A}) &= \sum_a \langle a| \otimes \mathbf{I}_{d'} \mathbf{A} |a\rangle \otimes \mathbf{I}_{d'} \\ &= \sum_{i,j,a} (\langle a| \otimes \mathbf{I}_{d'}) (|i\rangle \langle j| \otimes \mathbf{A}_{i,j}) (|a\rangle \otimes \mathbf{I}_{d'}) = \sum_a \mathbf{A}_{a,a} \\ \mathrm{Tr}_{\mathbb{C}^{d'}}(\mathbf{A}) &= \sum_a \mathbf{I}_d \otimes \langle a| \mathbf{A} \mathbf{I}_{d'} \otimes |a\rangle \\ &= \sum_{i,j,a} (\mathbf{I}_{d'} \otimes \langle a|) (|i\rangle \langle j| \otimes \mathbf{A}_{i,j}) (\mathbf{I}_{d'} \otimes |a\rangle) = \sum_{i,j} \mathrm{Tr} \mathbf{A}_{i,j} |i\rangle \langle j|\end{aligned}$$

Kraus' Theorem

Let $\mathcal{E} : \mathcal{B}(\mathbb{C}^d) \rightarrow \mathcal{B}(\mathbb{C}^{d'})$ be a completely positive map.

The following statements are equivalent:

1. Any Kraus representation of $\mathcal{E}$ is complete;
2. $\mathcal{E}$ has a complete Kraus representation;
3. $\mathcal{E}$ is trace preserving;
4. $\mathcal{E}$'s Choi matrix has $\text{Tr}_{\mathbb{C}^{d'}} \mathbf{J}_{\mathcal{E}} = \frac{\mathbf{I}_d}{d}$;

Proof: For any Kraus representation

$$\text{Tr}(\mathcal{E}(|i\rangle \langle j|)) = \text{Tr}\left(\sum_{k=0}^{m-1} \mathbf{K}_k |i\rangle \langle j| \mathbf{K}_k^\dagger\right) = \langle j| \sum_{k=0}^{m-1} \mathbf{K}_k^\dagger \mathbf{K}_k |i\rangle,$$

so if a Kraus representation is complete, all are, thus $1 \Leftrightarrow 2$. Since

$$\text{Tr}(\mathcal{E}(\rho)) = \text{Tr}\left(\sum_{i,j} \mathcal{E}(|i\rangle \langle i| \rho |j\rangle \langle j|)\right) = \sum_{i,j} \rho_{i,j} \text{Tr}(\mathcal{E}(|i\rangle \langle j|)),$$

so $\mathcal{E}$ is trace preserving iff $\text{Tr}(\mathcal{E}(|i\rangle \langle j|)) = \delta_{i,j} \forall i, j$ i.e. $2 \Leftrightarrow 3$.

Noting that $\mathcal{E}(|i\rangle \langle j|) = d(\langle i| \otimes \mathbf{I}_{d'}) \mathbf{J}_{\mathcal{E}} (|j\rangle \otimes \mathbf{I}_{d'})$, we find that the previous is equivalent to $\text{Tr}_{\mathbb{C}^{d'}} \mathbf{J}_{\mathcal{E}} = \frac{\mathbf{I}_d}{d}$, that is $3 \Leftrightarrow 4$.

Bistochastic Channel $\Leftrightarrow$ Unital Channel

$$\mathrm{Tr}(\mathbf{A}^\dagger \mathcal{E}(\mathbf{B})) = \mathrm{Tr}(\mathbf{A}^\dagger \sum_{k=0}^{m-1} \mathbf{K}_k \mathbf{B} \mathbf{K}_k^\dagger) = \mathrm{Tr}((\sum_{k=0}^{m-1} \mathbf{K}_k^\dagger \mathbf{A}^\dagger \mathbf{K}_k) \mathbf{B}) = \mathrm{Tr}((\sum_{k=0}^{m-1} \mathbf{K}_k^\dagger \mathbf{A} \mathbf{K}_k)^\dagger \mathbf{B})$$

So the adjoint of the linear map $\mathcal{E}$ s.t. $\mathcal{E}(\rho) = \sum_{k=1}^m \mathbf{K}_k \rho \mathbf{K}_k^\dagger$ is the linear map $\mathcal{E}^\dagger$ s.t. $\mathcal{E}^\dagger(\rho) = \sum_{k=1}^m \mathbf{K}_k^\dagger \rho \mathbf{K}_k$, so $\mathcal{E}$ is completely positive iff $\mathcal{E}^\dagger$ is completely positive.

However, $\mathcal{E}$ being trace preserving does not equate to $\mathcal{E}^\dagger$ being so. But, $\sum_{k=0}^{m-1} \mathbf{K}_k^\dagger \mathbf{K}_k = \mathcal{E}^\dagger(\mathbf{I}_{d'})$ so $\mathcal{E}$ is trace preserving iff $\mathcal{E}^\dagger$ is unital.

Like we said matrix $\mathbf{D}$ is bistochastic iff $\mathbf{D}, \mathbf{D}^\dagger$ are both stochastic, we say map $\mathcal{E}$ is bistochastic iff $\mathcal{E}, \mathcal{E}^\dagger$ are both channels, and we saw that this happens iff $\mathcal{E}$ is a unital quantum channel.

Stinespring's Dilation Theorem

We saw that adding an ancilla system E , performing a gate $\mathbf{U}$ and tracing out the ancilla E results in a quantum channel.

This theorem says that any channel $\mathcal{E}$ can be simulated in this way:
 $\mathcal{E}$ is a channel $\Leftrightarrow \mathcal{E}(\rho) = \text{Tr}_E (\mathbf{U}(|0\rangle_E \otimes \rho) \mathbf{U}^\dagger)$.

Consider a Kraus representation $\mathcal{E}(\rho) = \sum_{k=0}^{m-1} \mathbf{K}_k \rho \mathbf{K}_k^\dagger$ and notice that the matrix $\mathbf{V} = \sum_{k=0}^{m-1} |k\rangle \otimes \mathbf{K}_k \in \text{Mat}_{md',d}(\mathbb{C})$ has

$$\mathbf{V}^\dagger \mathbf{V} = \sum_{k,k'=0}^{m-1} \left(\langle k| \otimes \mathbf{K}_k^\dagger \right) \left(|k'\rangle \otimes \mathbf{K}_{k'} \right) = \sum_{k=0}^{m-1} \mathbf{K}_k^\dagger \mathbf{K}_k = \mathbf{I}_d$$

so $\mathbf{V}$ is an isometry, i.e. the columns of $\mathbf{V}$ are orthogonal vectors.

We can always extend this matrix up to an orthonormal basis using Gram-Schmidt, hence obtaining the columns of a gate $\mathbf{U}$.

Taking $d \mid md'$ (e.g. $m = dd'$), notice $\mathbf{U}(|0\rangle_E \otimes |\psi\rangle_{\mathbb{C}^d}) = \mathbf{V}|\psi\rangle$.

Measuring E gives k with probability $p_k = \text{Tr}(\mathbf{K}_k \rho \mathbf{K}_k^\dagger)$.

The post-measurement state is then $\rho_k = \mathbf{K}_k \rho \mathbf{K}_k^\dagger / p_k \otimes |k\rangle \langle k|_E$.
Tracing over E the result is $\mathcal{E}(\rho) = \sum_k p_k \mathbf{K}_k \rho \mathbf{K}_k^\dagger / p_k$, as desired.

Stinespring's Dilation Theorem

Let's look at what the algebra does for any $|\psi\rangle\langle\psi|$:

$$\begin{aligned}
 & \text{Tr}_E \left(\mathbf{U}(|0\rangle \otimes |\psi\rangle)(\langle 0| \otimes \langle\psi|)\mathbf{U}^\dagger \right) \\
 &= \text{Tr}_E \left(\begin{bmatrix} \mathbf{V} & ? \end{bmatrix} \begin{bmatrix} |\psi\rangle \\ \mathbf{0} \end{bmatrix} \begin{bmatrix} \langle\psi| & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{V}^\dagger \\ ? \end{bmatrix} \right) = \text{Tr}_E \left(\mathbf{V} |\psi\rangle\langle\psi| \mathbf{V}^\dagger \right) \\
 &= \text{Tr}_E \left(\begin{bmatrix} \mathbf{K}_0 \\ \mathbf{K}_1 \\ \vdots \\ \mathbf{K}_{m-1} \end{bmatrix} |\psi\rangle\langle\psi| \begin{bmatrix} \mathbf{K}_0^\dagger & \mathbf{K}_1^\dagger & \dots & \mathbf{K}_{m-1}^\dagger \end{bmatrix} \right) \\
 &= \text{Tr}_E \begin{bmatrix} \mathbf{K}_0 |\psi\rangle\langle\psi| \mathbf{K}_0^\dagger & \mathbf{K}_0 |\psi\rangle\langle\psi| \mathbf{K}_1^\dagger & \dots & \mathbf{K}_0 |\psi\rangle\langle\psi| \mathbf{K}_{m-1}^\dagger \\ \mathbf{K}_1 |\psi\rangle\langle\psi| \mathbf{K}_0^\dagger & \mathbf{K}_1 |\psi\rangle\langle\psi| \mathbf{K}_1^\dagger & \dots & \mathbf{K}_1 |\psi\rangle\langle\psi| \mathbf{K}_{m-1}^\dagger \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{K}_{m-1} |\psi\rangle\langle\psi| \mathbf{K}_0^\dagger & \mathbf{K}_{m-1} |\psi\rangle\langle\psi| \mathbf{K}_1^\dagger & \dots & \mathbf{K}_{m-1} |\psi\rangle\langle\psi| \mathbf{K}_{m-1}^\dagger \end{bmatrix} \\
 &= \sum_{k=0}^{m-1} \mathbf{K}_k |\psi\rangle\langle\psi| \mathbf{K}_k^\dagger.
 \end{aligned}$$

Positive Operator Valued Measures (POVMs)

Let $\mathcal{F}$ be Borel σ -algebra over Ω .

A POVM is a $F : \mathcal{F} \rightarrow \mathcal{B}(\mathcal{H})$ s.t. $F(\Omega) = \mathbf{I}_{\mathcal{H}}$ and for any $|\psi\rangle \in \mathcal{H}$, $\mu_{|\psi\rangle} : \mathcal{F} \rightarrow \mathbb{R}$ with $\mu_{|\psi\rangle}(E) = \langle\psi| F(E) |\psi\rangle$ is a probability measure

POVMs have the power to turn a finitely-dimensional $|\psi\rangle$ into an infinitely dimensional probability distribution. We won't do this.

For useful applications, we take Ω to be finite e.g. $\{1, 2, \dots, n\}$. Then all we need to specify is $\mathbf{F}_1 = F(\{1\}), \dots, \mathbf{F}_n = F(\{n\})$. For $\mu_{|\psi\rangle} \geq 0$ we need all $\mathbf{F}_i \geq 0$. By additivity of $\mu_{|\psi\rangle}$ we find

$$\mu_{|\psi\rangle}(E) = \sum_{i \in E} \mu_{|\psi\rangle}(\{i\}) = \sum_{i \in E} \langle\psi| F(\{i\}) |\psi\rangle = \langle\psi| \sum_{i \in E} \mathbf{F}_i |\psi\rangle$$

so $F(E) = \sum_{i \in E} \mathbf{F}_i$ and in particular $\mathbf{I}_{\mathcal{H}} = F(\Omega) = \sum_{i=1}^n \mathbf{F}_i$.

We identify a POVM with a collection $\{\mathbf{F}_i \geq 0\}$ s.t. $\sum_i \mathbf{F}_i = \mathbf{I}$.

Measuring a POVM upon ρ produces i with probability $\text{Tr}(\rho \mathbf{F}_i)$.

Naimark's Dilation Theorem

Prima facie, POVMs are more powerful than observables.
But actually, they can be implemented using gates and observables.

Let's take $\{\mathbf{F}_i \geq 0\}$ s.t. $\sum_{i=0}^{n-1} \mathbf{F}_i = \mathbf{I}_d$.

Using Cholesky decomposition we find $\mathbf{K}_i = \sqrt{\mathbf{F}_i}^\dagger$ s.t. $\mathbf{F}_i = \mathbf{K}_i^\dagger \mathbf{K}_i$.

So, $\mathcal{E}(\rho) = \sum_{i=0}^{n-1} \mathbf{K}_i \rho \mathbf{K}_i^\dagger$ is a quantum channel.

Construct now E , $\mathbf{U}$ as in the proof for Stinespring's theorem.

Note: if $n > dd'$ we don't take $m = dd'$ but $m = d\lceil n/d \rceil$.

Recall that measuring E we obtain i with the desired probability

$$p_i = \text{Tr}(\mathbf{K}_i \rho \mathbf{K}_i^\dagger) = \text{Tr}(\mathbf{K}_i^\dagger \mathbf{K}_i \rho) = \text{Tr}(\mathbf{F}_i \rho),$$

and the post-measurement state on the first subsystem is

$$\rho_i = \frac{\mathbf{K}_i \rho \mathbf{K}_i^\dagger}{\text{Tr} \mathbf{K}_i \rho \mathbf{K}_i^\dagger},$$

which notably depends on the choice of $\mathbf{K}_i$. If we took $\mathbf{K}'_i = \mathbf{V} \mathbf{K}_i$ for some unitary $\mathbf{V}$ the post-measurement state would be $\mathbf{V} \rho_i \mathbf{V}^\dagger$.

P vs CP

Up until now, we cared only for completely positive linear maps $\mathcal{E}$. As it turns out, positive maps are useful, but for different reasons. Applying it to halves of entangled states, they may turn negative.

Recall that $|\psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{K}_B$ is said to be separable iff it can be written as $|\psi\rangle_A = |\phi\rangle_A \otimes |\xi\rangle_B$, and entangled otherwise.

A density matrix ρ is said to be separable iff it corresponds to a mixture of separable states, and entangled otherwise.

Any positive and trace-preserving linear map $\mathcal{E} : \mathcal{B}(\mathbb{C}^d) \rightarrow \mathcal{B}(\mathbb{C}^{d'})$ has its extensions map separable states to valid (separable) states,

$$(\mathbf{I}_n \otimes \mathcal{E}) \left(\sum_i p_i |\phi_i\rangle \langle \phi_i| \otimes |\psi_i\rangle \langle \psi_i| \right) = \sum_i p_i |\phi_i\rangle \langle \phi_i| \otimes \mathcal{E}(|\psi_i\rangle \langle \psi_i|),$$

but if $\mathcal{E}$ is not n -positive, then (by definition) there must be some ρ such that $(\mathbf{I}_n \otimes \mathcal{E})\rho \not\geq 0$, and by the above, this ρ is not separable.

Entanglement Witnesses

Notice now that separable states form a convex set within the real Hilbert space of Hermitian matrices acting on $\mathbb{C}^n \otimes \mathbb{C}^d$.

It follows that, conversely, if ρ is entangled then there is a separating hyperplane between ρ and the set of separable states.

This takes the form of an observable $\mathbf{O}$ such that

$$\text{Tr } \mathbf{O} \rho < 0; \quad \text{Tr } \mathbf{O} |\phi\rangle \langle \phi| \otimes |\psi\rangle \langle \psi| \geq 0 \quad \forall |\phi\rangle \in \mathbb{C}^n, |\psi\rangle \in \mathbb{C}^d,$$

something known as an entanglement witness.

We conclude that for every entangled state there's an entanglement witness that separates it from the set of separable states.

PnCP Separability Criterion

We interpret $\mathbf{O}$ as the (corner of the) Choi matrix of a map $\mathcal{E}$.

Since $\mathbf{O} = \mathbf{O}^\dagger$, we find $\mathcal{E}(|i\rangle\langle j|) = \mathcal{E}(|j\rangle\langle i|)^\dagger$ so $\mathcal{E}(\mathbf{Z}_a) = \mathcal{E}(\mathbf{Z}_a)^\dagger$, $\mathcal{E}(\mathbf{X}_{b,c}) = \mathcal{E}(\mathbf{X}_{b,c})^\dagger$ and $\mathcal{E}(\mathbf{Y}_{d,e}) = \mathcal{E}(\mathbf{Y}_{d,e})^\dagger$ thus $\mathcal{E}$ maps Hermitian matrices to Hermitian matrices.

We can actually show that this $\mathcal{E}$ is a positive map.

If there is $|\psi_0\rangle$ s.t. $\mathcal{E}(|\psi_0\rangle\langle\psi_0|) \not\geq 0$, then, since it is Hermitian, it must have an eigenvector $|\phi\rangle$ with $\langle\phi|\mathcal{E}(|\psi_0\rangle\langle\psi_0|)|\phi\rangle < 0$ so

$$\begin{aligned} & \text{Tr}((\mathbf{I}_d \otimes \mathcal{E})|\beta_{00}\rangle\langle\beta_{00}| |\psi_0\rangle\langle\psi_0| \otimes |\phi\rangle\langle\phi|) \\ &= \frac{1}{d} \text{Tr}((\mathbf{I}_d \otimes \mathcal{E}) \left(\sum_{i,j} |\psi_j\rangle\langle\psi_i| \otimes |\psi_i\rangle\langle\psi_j| \right) |\psi_0\rangle\langle\psi_0| \otimes |\phi\rangle\langle\phi|) \\ &= \frac{1}{d} \text{Tr} \left(\sum_{i,j} \langle\psi_0|\psi_j\rangle\langle\psi_i|\psi_0\rangle \otimes \langle\phi|\mathcal{E}(|\psi_i\rangle\langle\psi_j|)|\phi\rangle \right) \\ &= \frac{\langle\phi|\mathcal{E}(|\psi_0\rangle\langle\psi_0|)|\phi\rangle}{d} < 0, \end{aligned}$$

contradicting the hyperplane definition of $\mathbf{O}$. Thus $\mathcal{E}$ is positive.

So $\rho \in \mathcal{B}(\mathbb{C}^n \otimes \mathbb{C}^d)$, $n \leq d$, is entangled iff there is a positive map $\mathcal{E} : \mathcal{B}(\mathbb{C}^d) \rightarrow \mathcal{B}(\mathbb{C}^d)$ s.t. $\mathbf{I}_n \otimes \mathcal{E}(\rho) \not\geq 0$. This $\mathcal{E}$ is not CP.

Positive Partial Transpose (PPT) Criterion

For example, $\mathcal{T}(\rho) = \rho^T$ is positive but not completely positive as

$$(\mathbf{I}_d \otimes \mathcal{T}) |\beta_{00}\rangle \langle \beta_{00}| \underbrace{(|01\rangle - |10\rangle)}_{|\psi\rangle} = \frac{1}{d} \sum_{i,j} |i\rangle \langle j| \otimes |j\rangle \langle i| |\psi\rangle = -|\psi\rangle.$$

The map $\mathbf{I}_n \otimes \mathcal{T}$ defined by $\mathbf{I}_n \otimes \mathcal{T}(|a\rangle \langle b| \otimes |i\rangle \langle j|) = |a\rangle \langle b| \otimes |j\rangle \langle i|$ is called the partial transpose. As discussed, for any separable $\rho = \sum_i p_i |\phi_i\rangle \langle \phi_i| \otimes |\psi_i\rangle \langle \psi_i|$ we will have $\mathbf{I}_n \otimes \mathcal{T}(\rho) \geq 0$ so, **if $\rho \in \mathcal{B}(\mathbb{C}^n \otimes \mathbb{C}^d)$ has $\mathbf{I}_n \otimes \mathcal{T}(\rho) \not\geq 0$ then ρ is entangled.**

The PPT criterion (also known as the Peres-Horodecki criterion) is both necessary and sufficient iff $nd \leq 6$, i.e. for qubit-qubit and qubit-qutrit entanglement detection.

$$\mathbf{I}_n \otimes \mathcal{T} \left(\begin{bmatrix} \mathbf{A}_{1,1} & \cdots & \mathbf{A}_{1,n} \\ \vdots & \ddots & \vdots \\ \mathbf{A}_{n,1} & \cdots & \mathbf{A}_{n,n} \end{bmatrix} \right) = \begin{bmatrix} \mathbf{A}_{1,1}^T & \cdots & \mathbf{A}_{1,n}^T \\ \vdots & \ddots & \vdots \\ \mathbf{A}_{n,1}^T & \cdots & \mathbf{A}_{n,n}^T \end{bmatrix}.$$

Does **O** witness the entanglement of $\rho \in \mathcal{B}(\mathbb{C}^{2 \times 2})$?

$$\rho = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\dagger & \mathbf{C} \end{bmatrix} \Rightarrow \mathbf{I}_2 \otimes \mathcal{T}(\rho) = \begin{bmatrix} \mathbf{A}^\top & \mathbf{B}^\top \\ \overline{\mathbf{B}} & \mathbf{C} \end{bmatrix}$$

Checklist:

1. Is ρ a valid density matrix? ($\rho = \rho^\dagger$, $\rho \geq 0$, $\text{Tr } \rho = 1$)
2. Is **O** a valid observable? (**O** = **O**[†])
3. Is ρ entangled? (PPT: $\mathbf{I}_2 \otimes \mathcal{T}(\rho) \not\geq 0$)
4. Can **O** be a witness? (**O** $\not\geq 0$)
5. Is **O** a true witness? (Does it correspond to PnCP $\mathcal{E}$?)

Setting **O** = **J** _{$\mathcal{E}$} , is $\mathcal{E} \geq 0$? ($\mathcal{E}(\mathbf{X}) \geq 0 \forall \mathbf{X} \geq 0$)

Note: since $\mathcal{E}$ is linear, the convex wrapper of the image of the positive cone is the image of the convex hull of the positive cone, i.e. this is equivalent to $\mathcal{E}(|\psi\rangle\langle\psi|) \geq 0$ for all pure $|\psi\rangle$, which we know we can parametrize as

$|\psi\rangle = \sqrt{1-p}|0\rangle + \sqrt{p}e^{2\pi i\theta}|1\rangle$ where $0 \leq p \leq 1$, $0 \leq \theta \leq 1$.

6. Finally, does **O** witness the entanglement of ρ ? ($\text{Tr}(\mathbf{O}\rho) < 0$)

$$\mathbf{J}_{\mathcal{E}} = \mathbf{O} = \begin{bmatrix} \mathbf{M} & \mathbf{N} \\ \mathbf{N}^\dagger & \mathbf{P} \end{bmatrix} \Rightarrow \mathcal{E}\left(\begin{bmatrix} x & y \\ z & t \end{bmatrix}\right) = 2x\mathbf{M} + 2y\mathbf{N} + 2z\mathbf{N}^\dagger + 2t\mathbf{P}$$

Follow-up Reading

- ▶ J.Watrous – The theory of quantum information
Cambridge university press (2018)
- ▶ E.Chitambar, G.Gour – Quantum resource theories
Reviews of modern physics, 91(2), 025001 (2019)